

SwasthyaSetu AI

An offline-first, ₹2500 edge-AI health screening platform for community health workers.

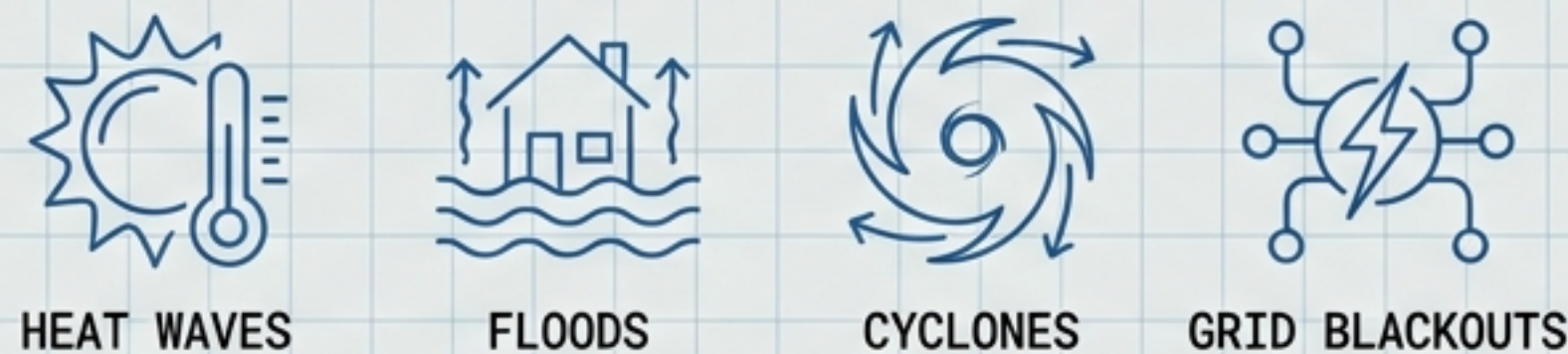
Core Value Statement:

Record vitals → Get deterministic triage → Understand in plain language → Act fast. All without internet.



The Reality of Healthcare in Climate Emergencies

The Triggers



Extreme weather and infrastructure failure.

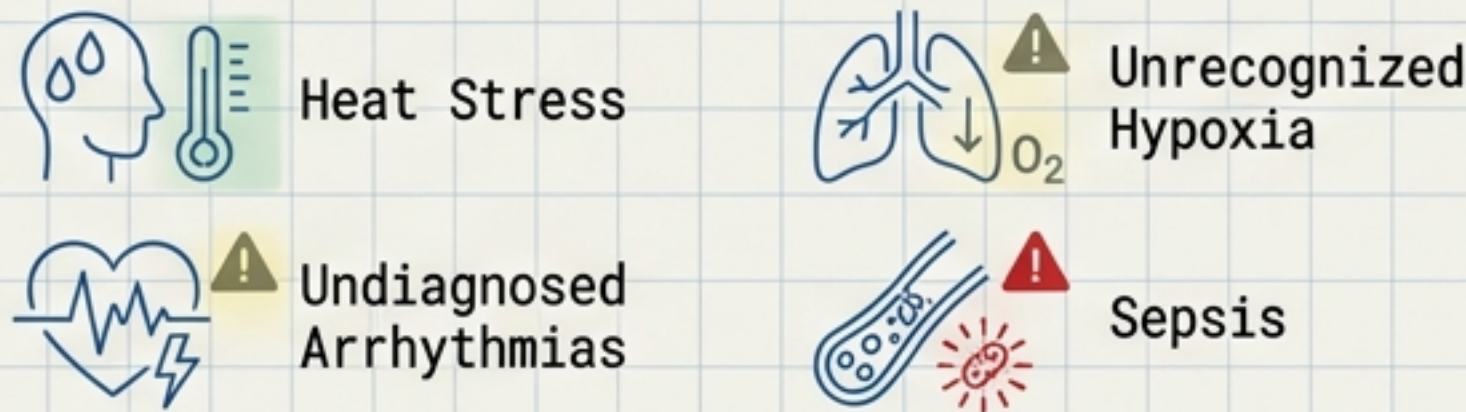
The Vulnerable

Focus on rural populations, elderly citizens, and outdoor workers who cannot travel to centralized clinics.



The Clinical Risks

Highlighting life-threatening conditions like heat stress, unrecognized hypoxia, undiagnosed arrhythmias, and sepsis.



The Core Challenge

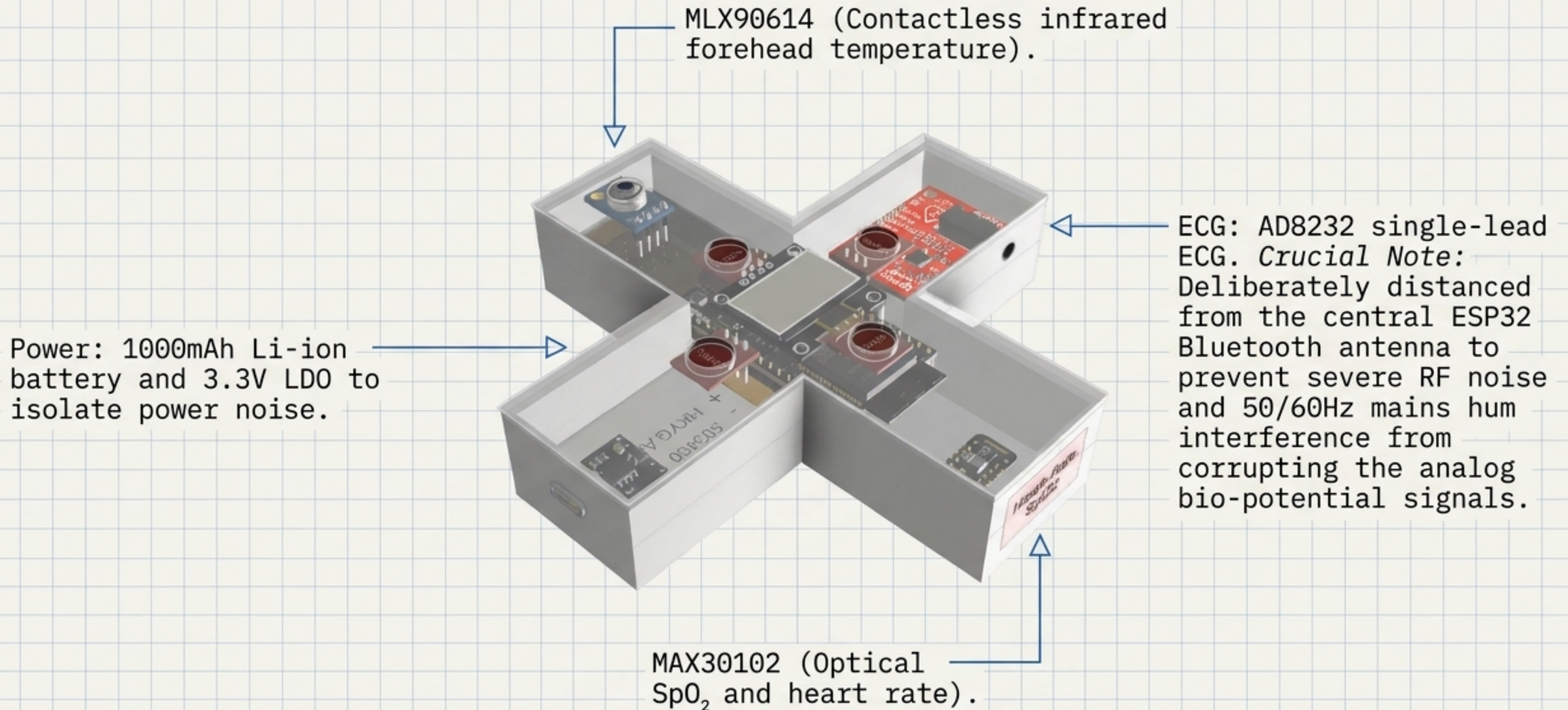
Community health workers lack the tools to perform perform clinical-grade triage when the cloud is inaccessible and medical literacy is a barrier.



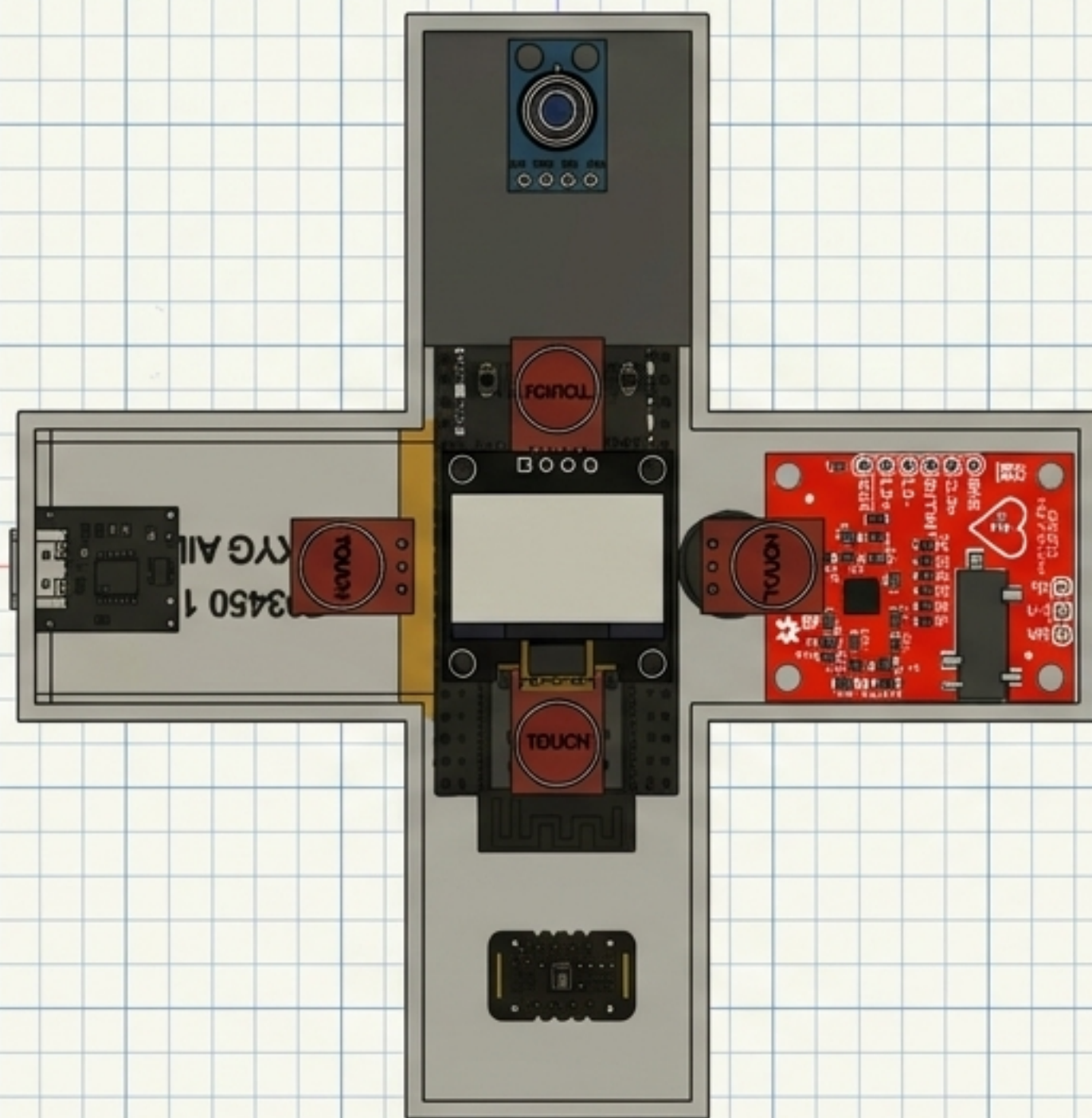
The SwasthyaSetu Paradigm

Dimension	Traditional	SwasthyaSetu AI
Infrastructure Dependency	Cloud-dependent; fails during blackouts.	100% Offline; utilizes bundled MBTiles and SMS SOS payloads.
Medical Logic & AI	Diagnostic guessing with high LLM hallucination risk.	Deterministic mathematical triage with one-way provenance (AI is restricted strictly to explanations).
Hardware Form Factor	Expensive, fragile, centralized medical equipment.	₹2500 ruggedized, plus-shaped 3D-printed edge device.

Hardware Form Factor: Intentional Physical Isolation



Component Mapping & Economic Viability (Total Cost: ₹2500)



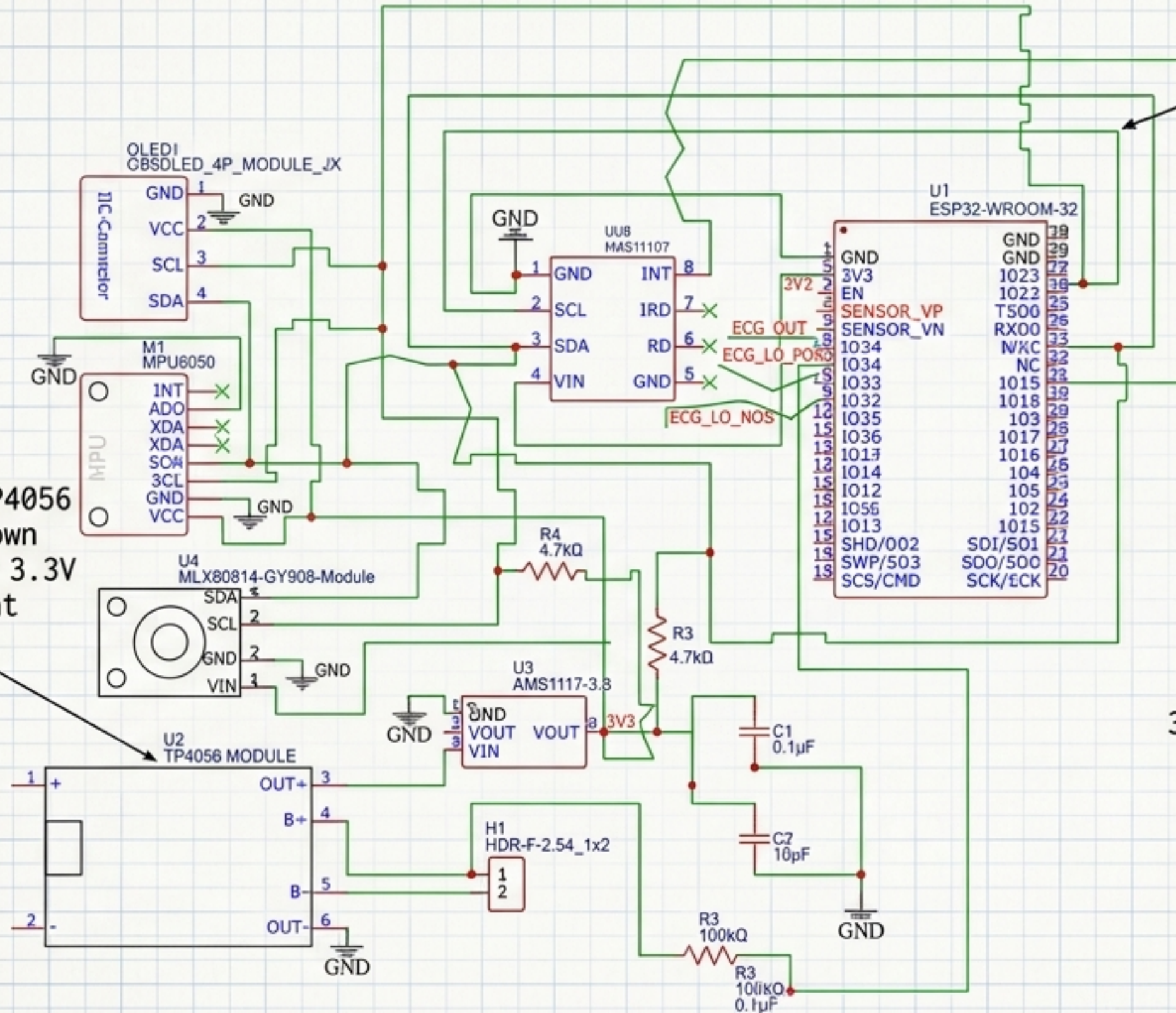
	Group	Component	Cost (₹)
1	Processing	ESP32-WROOM-32 DevKit	₹330
		SSD1306 OLED	₹150
2	Sensors	MAX30102 Breakout	₹300
		AD8232 Breakout	₹500
		MLX90614	₹400
3	Power & Casing	1000mAh Li-ion + TP4056 + AMS1117	₹400
		3D Enclosure/Electrodes	₹420

The reusable ecosystem—the one-time ₹2500 CapEx utilizes standard, cheap disposable Ag/AgCl snap electrodes for patient hygiene.

Circuit Architecture & Signal Integrity

1. Power Stepping:

Li-ion (3.7V) → TP4056
→ AMS1117 steps down
to a highly stable 3.3V
rail (vital for flat
ECG baselines).

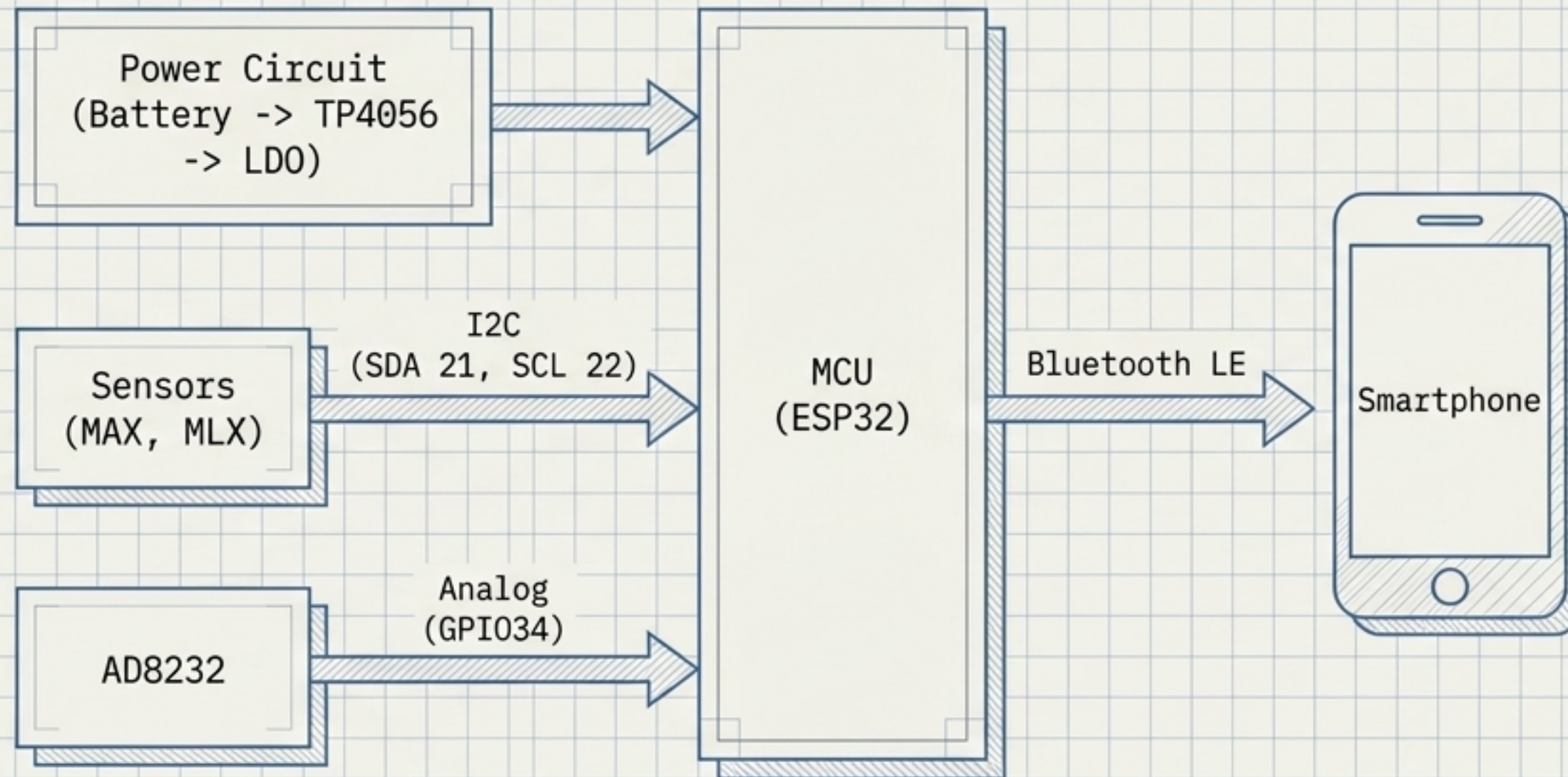


2 **I2C Bus Constraints:** ESP32 internal pull-ups are too weak for 3 parallel devices (OLED, MAX, MLX). External 4.7kΩ pull-ups added on SDA (GPIO21) and SCL (GPIO22).



3. **Analog Lead-Off Detection:** AD8232 LO+ (GPIO32) and LO- (GPIO33) are equipped with 10kΩ pull-downs to prevent floating states from triggering false electrode disconnections.

Firmware Execution & The BLE Bridge



Execution Logic Panel

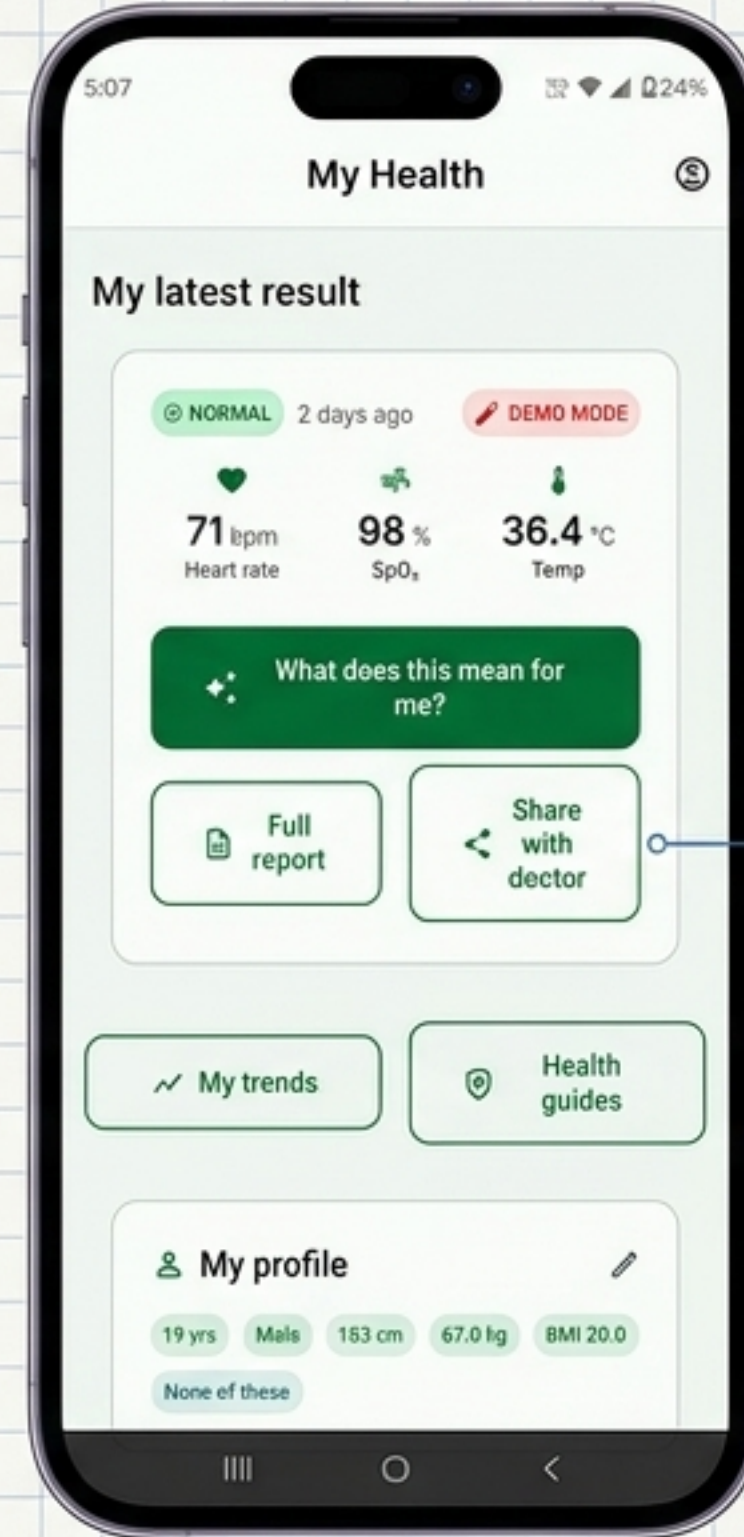
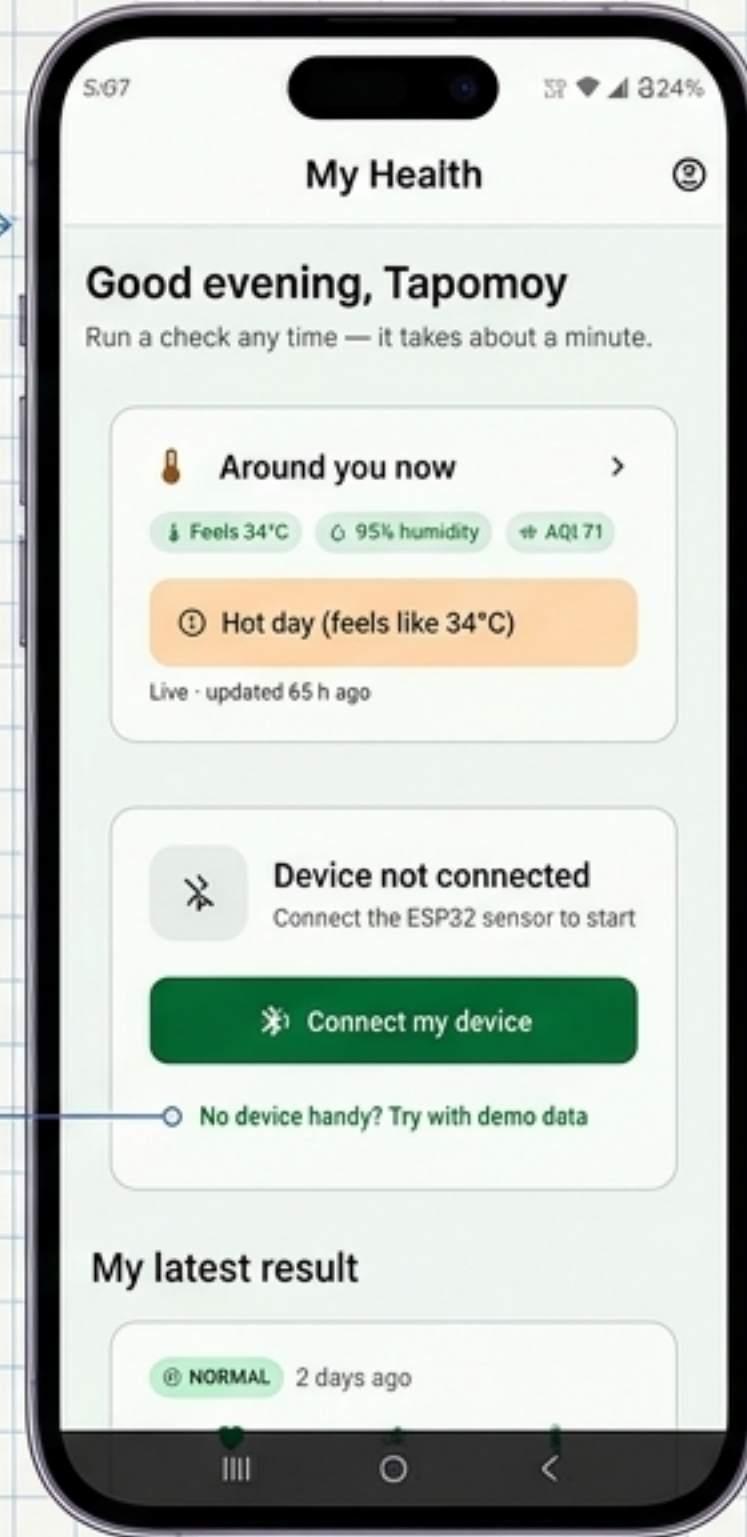
- **Resilience:** Hardware timers poll the ECG ADC every 20ms for continuous waveforms.
- **Telemetry:** I2C sensors are polled every 1 second.
- **Honesty Link State:** BLE advertises automatically; if the link drops, the app intercepts and preserves waveform data.

The App Ecosystem: Zero-Dependency & Clinical Honesty

Flutter 3.47+ Architecture:
A highly responsive, totally
offline interface.

Demo Mode Fallback: If a
health worker lacks the
physical ₹2500 hardware,
they can still explore the
app's guidelines.

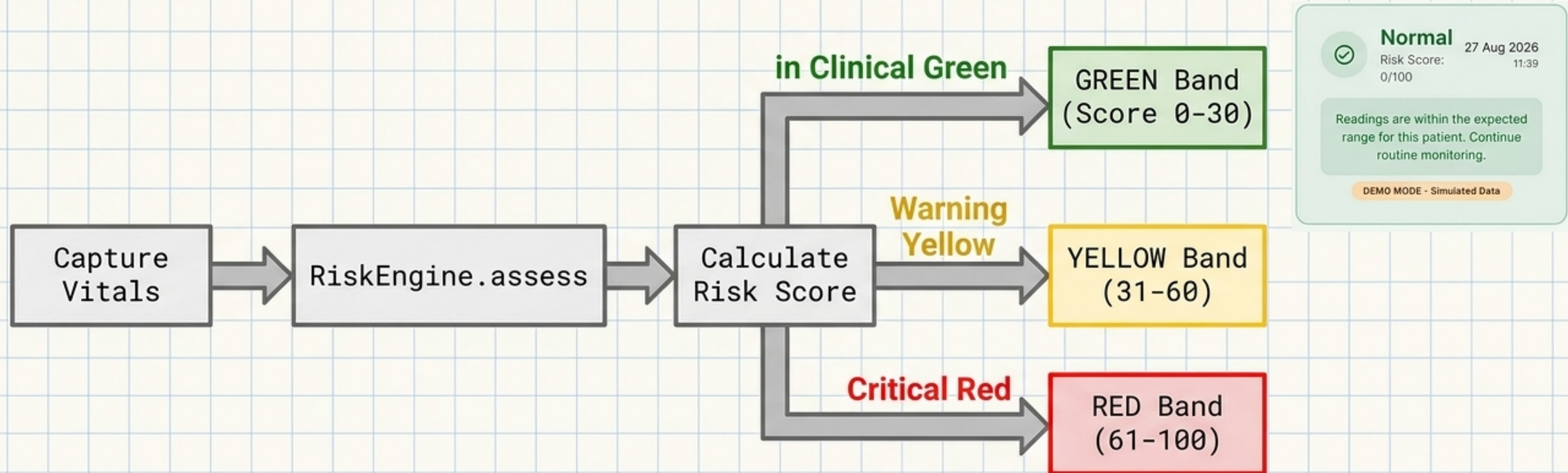
Simulated readings are
permanently flagged with a
prominent **DEMO MODE** badge
to prevent masquerading as
clinical vitals.



Consent-Gated Sync:
No automatic background
uploads. Location and sync
permissions are fully
optional and off by
default.

The Safety Boundary: Deterministic Triage Engine

AI IS STRICTLY FORBIDDEN FROM CLINICAL DECISIONS. The Risk Engine is purely **mathematical**. Raw vitals are evaluated against dynamic **WHO/ICMR** thresholds (adjusted for patient vulnerabilities) to hard-code the risk band before any AI is involved.



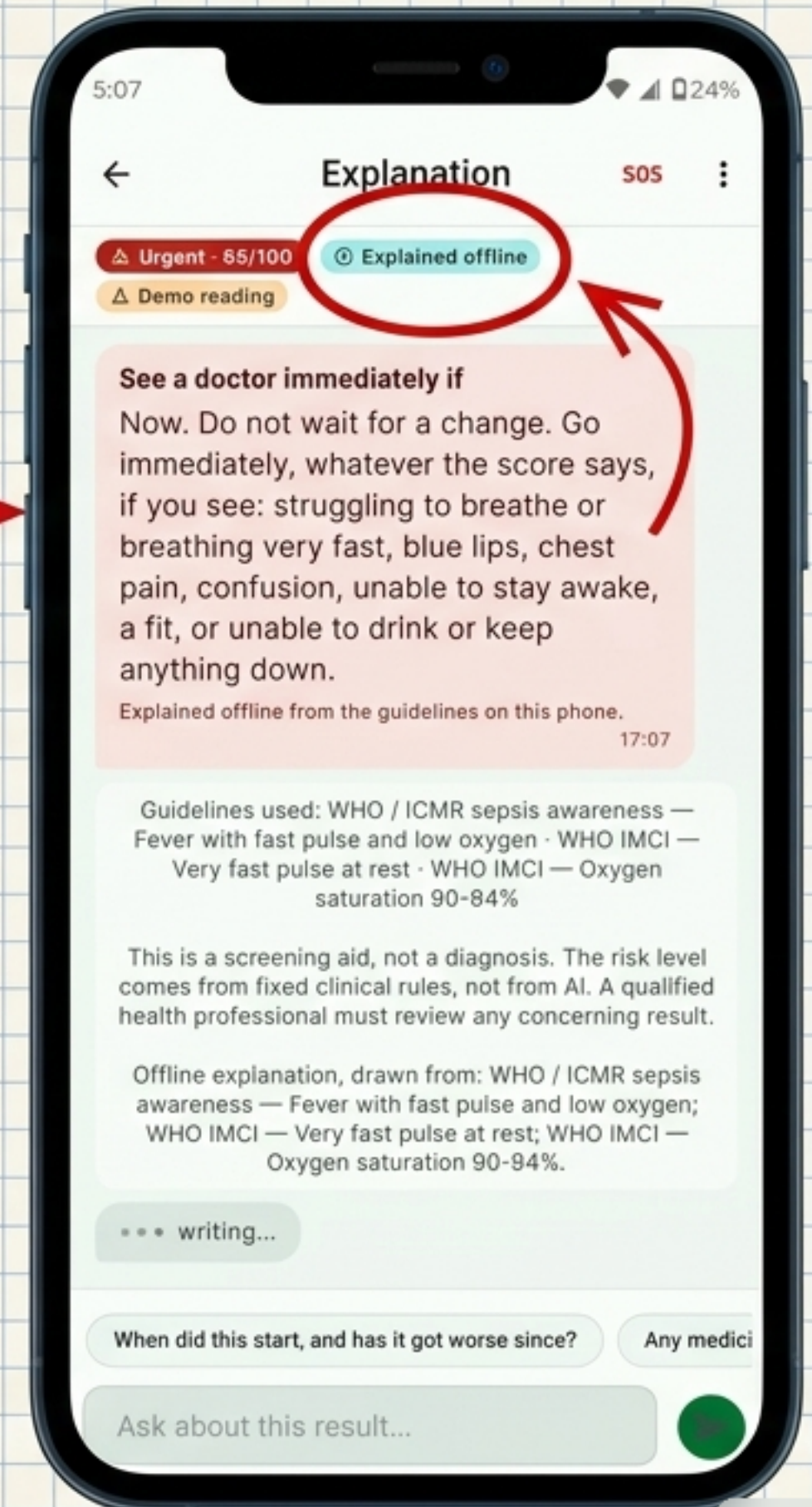
Two-Tier Explanation Architecture

Tier 1 (Online - Gemini Flash)

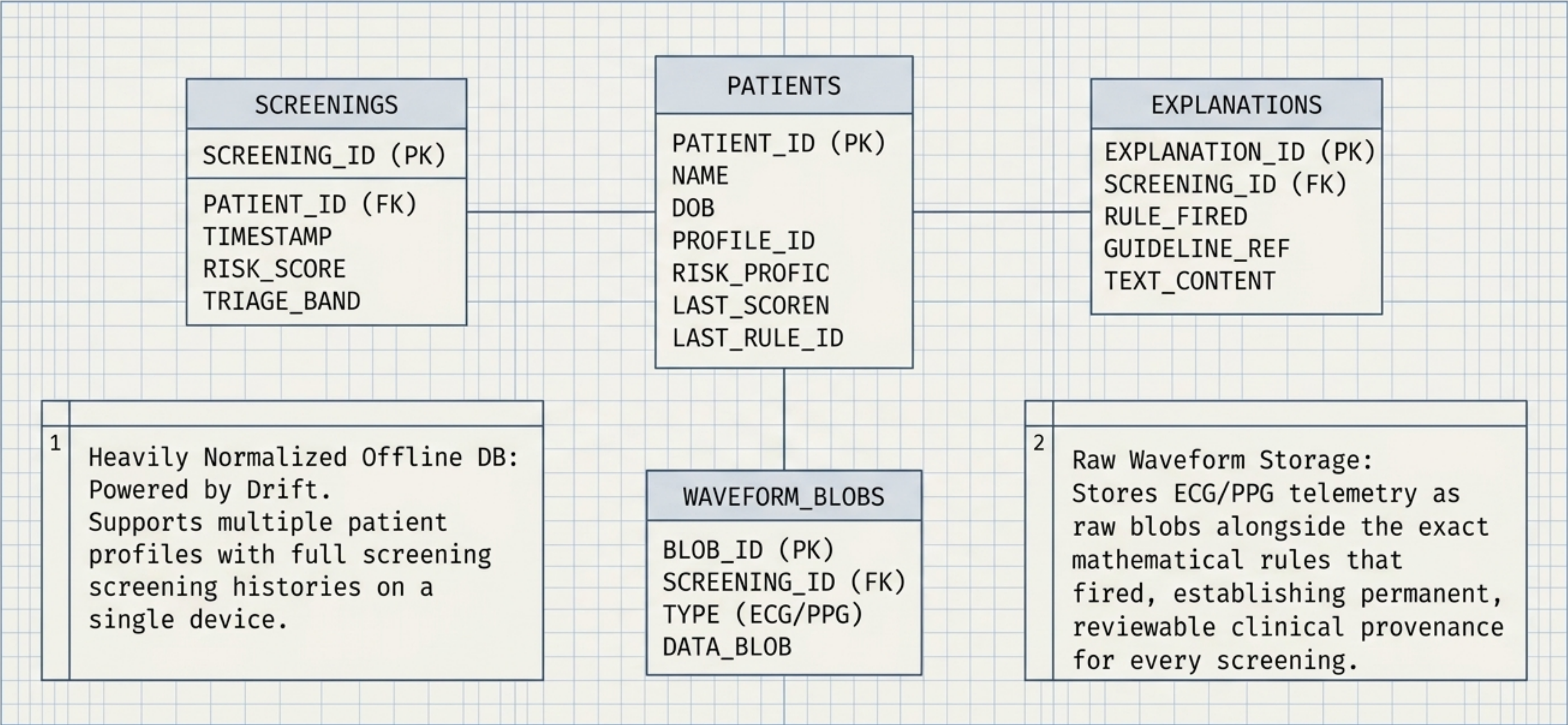
A lightweight LLM prompted with a strict JSON constraint. It is forced to accept the deterministic score, summarize findings in plain language, and suggest follow-up questions. Hallucinated diagnoses are structurally banned.

Tier 2 (Offline Fallback - Local Corpus)

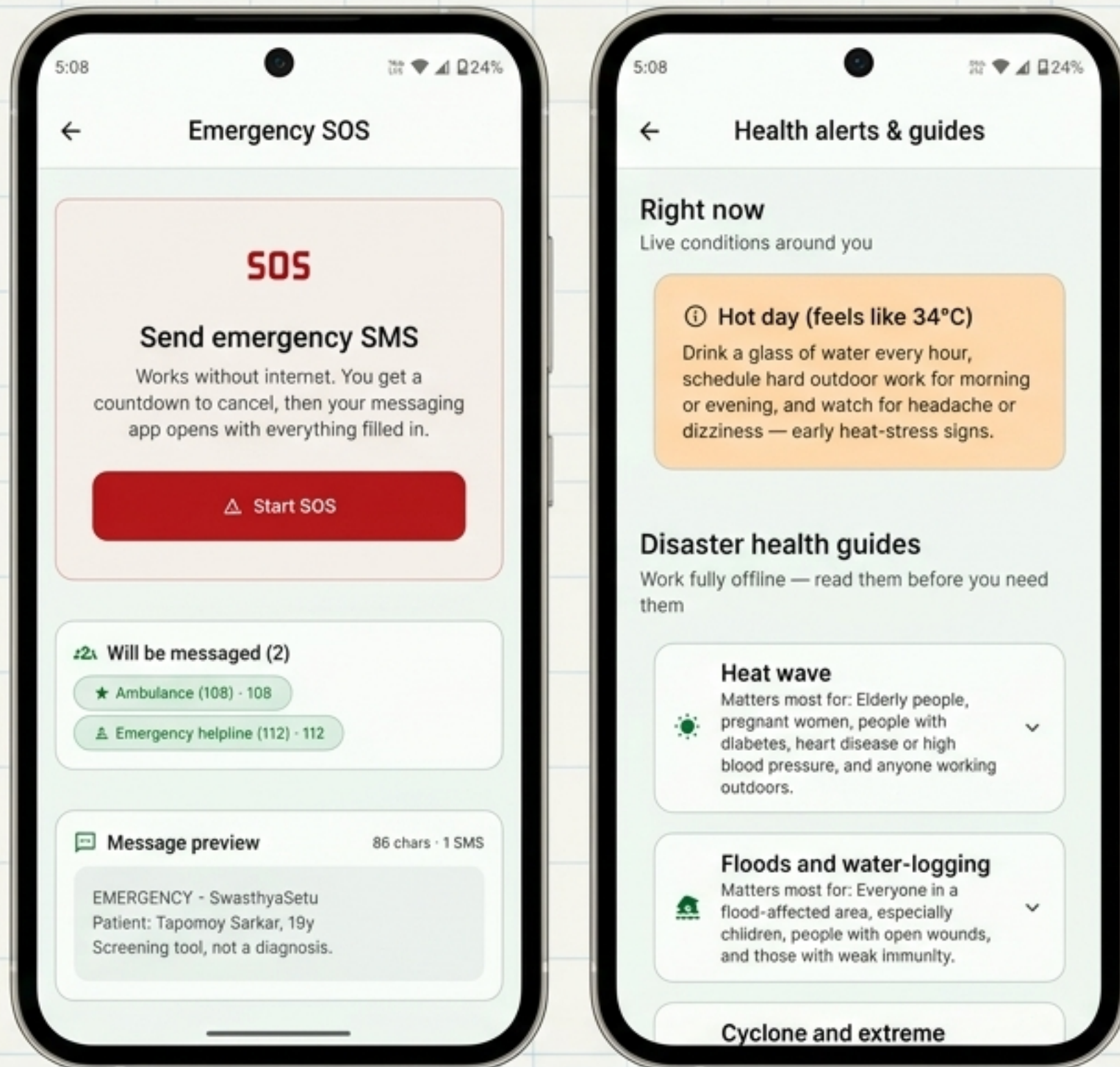
The true innovation for disaster scenarios. If connectivity drops, the app intercepts the failure, queries a bundled SQLite database of WHO/ICMR clinical guidelines, and extracts paragraphs matching the fired rules.



Edge Persistence: Drift SQLite & Data Provenance



Emergency Action: SMS SOS & Bundled Maps



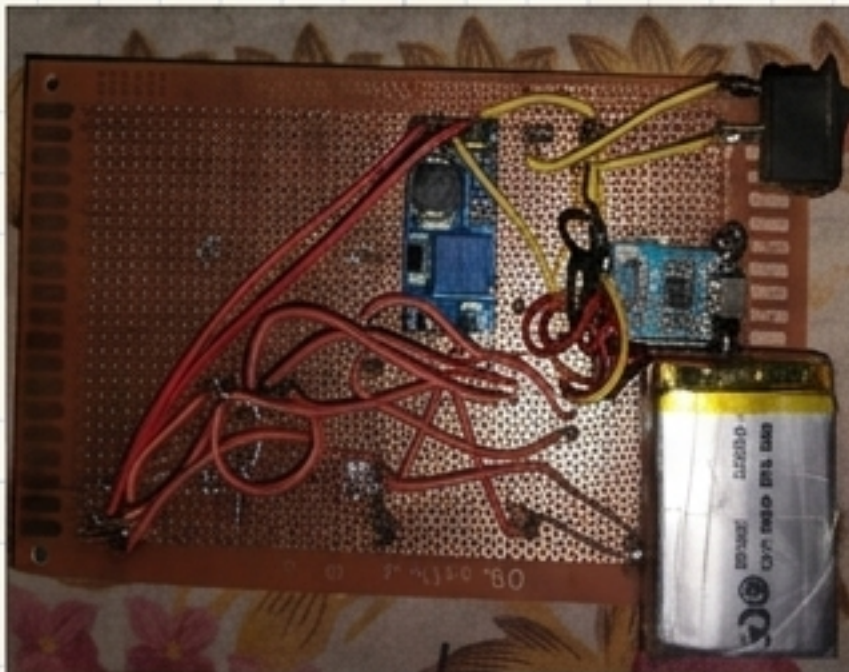
- **SMS Payload Parsing:** Bypasses the need for internet data by composing a dense SMS containing triage results + coordinates, launching the native messaging app.
- **Offline Map Architecture:** Utilizes real OpenStreetMap data via bundled .mbtiles, allowing geographical tagging without a single byte of cellular data.
- **Fall Detection:** Combines smartphone accelerometers with the sensor board IMU to detect patient collapse.

Technical Prototyping & Live Validation

Front



Back



Proof of Work Checklist

- **Firmware Self-Test:** Successfully passes internal tests 1-11 (I2C scan, AD8232 signal range, BLE TX).
- **Signal Quality:** Achieves stable baselines with >10 dB SNR, clear R-peaks, and $0.5\text{--}2$ mV amplitude.
- **Engineering Rigor:** Backed by 428 automated unit/widget tests in Flutter, ensuring the deterministic logic never fails in production.

Feasibility & Risk Mitigation

Medical Liability	Hardware Fragility	Total Internet Blackout
Risk: AI hallucinates a fatal diagnosis.	Risk: Devices break in the field.	Risk: Cloud dependencies brick the app during a cyclone.
Fixed Strategy: AI is strictly quarantined to generating explanations. Triage relies purely on fixed clinical deterministic rules.	Fixed Strategy: The 125x125mm 3D-printed enclosure provides physical protection, protection, and the reusable TP4056 Li-ion ecosystem eliminates recurring battery costs.	Fixed Strategy: Bundled SQLite clinical corpus and MBTiles guarantee 100% core functionality with zero bytes of data.

Empowering the Frontline

Socio-Economic Impact

Early, decentralized detection of Sepsis, Hypoxia, and Arrhythmias drastically reduces late-stage ICU admissions. Converts ₹2500 CapEx into thousands of saved hospitalizations.

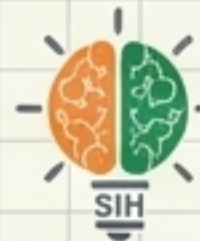
Environmental Impact

Zero-cloud-compute carbon footprint for routine screenings. E-waste minimized via reusable hardware and standard disposable Ag/AgCl pads.

The Bottom Line: SwasthyaSetu removes the burden of clinical interpretation from community health workers, converting raw vitals into clear action steps.

Inspect the Code & Hardware Architecture:

<https://github.com/krishnendukoley2007-arch/swasthyasetu-ai>



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